

18/06/2026 12 CBSE Ibn Math MATRICES Question Paper[View Question IDs](#)**18/06/2026 12 CBSE Ibn Math MATRICES****Max.Marks: 40.0****Duration: 60Minutes****Date: 18/06/26****Instructions**

Web Instructions

TEST**1. General Instructions:**

Read all instructions carefully before attempting the questions.

Write your name, class, and date at the top of the answer sheet/test paper.

Attempt all questions unless instructed otherwise. All questions are compulsory unless specified.

Write neatly and legibly. Answers that are unclear may not be awarded marks.

Manage your time wisely. Allocate time to each section according to the marks assigned.

Do not use unfair means. Any form of cheating will result in disqualification of the test.

Use only blue or black pen for writing answers. Pencils may be used for diagrams or rough work.

Check your paper thoroughly before submission to ensure all questions are answered.

Submit the paper promptly on portal as explained

All the best! Do your best.

If $A = [a_{ij}]_{3 \times 4}$ is matrix given by $A = \begin{bmatrix} 4 & -2 & 1 & 3 \\ 5 & 7 & 9 & 6 \\ 21 & 15 & 18 & -25 \end{bmatrix}$. Then, $a_{23} + a_{24}$ will be equal to the element [1]

1) a_{14}

2) a_{13}

3) a_{32}

4) a_{44}

Marks : 1.0

Negative Marks :

Answer Key:

SOLUTION:

2. Assume X, Y, Z, W and P are matrices of order $2 \times n$, $3 \times k$, $2 \times p$, $n \times 3$ and $p \times k$, respectively.

The restriction on n, k and p so that $PY + WY$ will be defined are [1]

1) $k = 3, p = n$

2) $k = 2, p = 3$

3) k is arbitrary, $p = 2$

4) p is arbitrary, $k = 3$

Marks : 1.0

Negative Marks :

Answer Key:

SOLUTION:

3. If A and B are any two square matrices of the same order, then [1]

1) $\text{adj}(AB) = \text{adj}(A) \text{adj}(B)$

2) $(AB)^t = B^t A^t$

3) $AB = O$

4) $(AB)^t = A^t B^t$

Marks : 1.0

Negative Marks :

Answer Key:

SOLUTION:

4. If $A = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$, then [1]

1) $AB = BA$

2) $AB \neq BA$

3) BA is not defined

4) AB is not defined

Marks : 1.0

Negative Marks :

Answer Key:

SOLUTION:

5. If $A = \begin{vmatrix} p & q \\ 0 & 1 \end{vmatrix}$, then A^5 is: [1]

1) $\begin{vmatrix} p^5 & q\left(\frac{p^5-1}{p-1}\right) \\ 0 & 1 \end{vmatrix}$

2) $\begin{vmatrix} p^5 & \left(\frac{p^5-1}{p-1}\right) \\ 0 & 1 \end{vmatrix}$

3) $\begin{vmatrix} p^5 & q^5 \\ 0 & 1 \end{vmatrix}$

4) $\begin{vmatrix} p^4 & q\left(\frac{p^5-1}{p-1}\right) \\ 1 & 10 \end{vmatrix}$

Marks : 1.0

Negative Marks :

Answer Key:

SOLUTION:

6.

Match the column:

$A = \begin{bmatrix} -3 & 1 \\ -2 & 4 \\ 5 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -3 \\ 0 & -2 \\ -2 & 4 \end{bmatrix}$ then, the (a) If order of matrix $3A - 2B$ is	(i) 3×3
(b) The order of the matrix $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} [2 \ 3 \ 4]$ is	(ii) 4×4
(c) If a matrix has 3 elements then a possible order it can have is	(iii) 3×2
(d) If A is a matrix of order 4×3 and B is a matrix of order 3×4 , then the order of matrix AB is	(iv) 1×3

[2]

Marks : 2.0

Negative Marks :

Answer Key:

SOLUTION:

7. Match the column:

If A and B are matrices of order $m \times n$, then

(a) $k(A + B) =$	(i) $-(kA) = k(-A)$
(b) $(k^1 + k^2)A =$	(ii) $kA + kB$
(c) $k^1 k^2 A =$	(iii) $k^1 A + k^2 A$
(d) $(-k)A =$	(iv) $k^1 (k^2 A) = k^2 (k^1 A)$

[2]

Marks : 2.0

Negative Marks :

Answer Key:

SOLUTION:

8.

Match the column:

(a) A matrix having only one row and any number of columns is called a	(i) Rectangular Matrix
(b) A matrix having only one column and any number of rows is called a	(ii) Row Matrix
(c) A matrix of order $m \times n$, such that $m \neq n$, is called	(iii) null/zero matrix
(d) A matrix of any order, having all its elements are zero, is called a	(iv) Column Matrix

[2]

Marks : 2.0

Negative Marks :

Answer Key:

SOLUTION:

9.

Match the column:

(a) If A and B are symmetric matrices of the same order, then the product AB is symmetric, iff	(i) Skew-symmetric
(b) Every square matrix can be expressed uniquely as the sum of a	(ii) Symmetric
(c) If $A^T = -A$, the matrix is	(iii) $BA = AB$
(d) All positive integral powers of a symmetric matrix are	(iv) Symmetric and a skew-symmetric matrix

[2]

Marks : 2.0

Negative Marks :

Answer Key:

SOLUTION:

10.

Match the column:

(a) $(A + B)^T =$	(i) kA^T
(b) $(AB)^T =$	(ii) $A^T + B^T$
(c) $(kA)^T =$	(iii) $(A^T)^N$
(d) $(A^N)^T =$	(iv) $B^T A^T$

[2]

Marks : 2.0

Negative Marks :

Answer Key:

SOLUTION:

11. If $A = \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix}$ & $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$; Find k, So that $A^2 = kA - 2I$. [3]

Marks : 3.0

Negative Marks :

Answer Key:

SOLUTION:

12. Let $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$ and $f(x) = x^2 - 4x + 7$. Show that $f(A) = O$. Use this result to find A^5 [3]

Marks : 3.0

Negative Marks :

Answer Key:

SOLUTION:

13. Construct a 4×3 matrix $A = [a_{ij}]$ whose elements a_{ij} are given by:
 $a_{ij} = 2i + \frac{i}{j}$ [3]

Marks : 3.0

Negative Marks :

Answer Key:

SOLUTION:

14. Find X and Y, if $2X + 3Y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix}$ and $3X + 2Y = \begin{bmatrix} 2 & -2 \\ -1 & 5 \end{bmatrix}$ [3]

Marks : 3.0

Negative Marks :

Answer Key:

SOLUTION:

15. If $\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$, find the value of b. [3]

Marks : 3.0

Negative Marks :

Answer Key:

SOLUTION:

16. Let $A = \begin{bmatrix} 3 & 2 & 7 \\ 1 & 4 & 3 \\ -2 & 5 & 8 \end{bmatrix}$. Find matrices X and Y such that $X + Y = A$, where X is a symmetric and Y is a skew - symmetric matrix. [5]

Marks : 5.0

Negative Marks :

Answer Key:

SOLUTION:

17. If $A = \begin{bmatrix} 5 & -1 & 4 \\ 2 & 3 & 5 \\ 5 & -2 & 6 \end{bmatrix}$, find A^{-1} and use it to solve the following system of equations:

$$5x - y + 4z = 5$$

$$2x + 3y + 5z = 2$$

$$5x - 2y + 6z = -1 \quad p[5]$$

Marks : 5.0

Negative Marks :

Answer Key:

SOLUTION:

Solution

12 CBSE MATH MATRICES

Class 12 - Mathematics

1.

(c) a_{32}

Explanation:

$$\text{The given matrix is } A = \begin{bmatrix} 4 & -2 & 1 & 3 \\ 5 & 7 & 9 & 6 \\ 21 & 15 & 18 & -25 \end{bmatrix}$$

Here, $a_{23} = 9$ and $a_{24} = 6$

$$\therefore a_{23} + a_{24} = 9 + 6 = 15$$

Also, 15 lies in 3rd row and 2nd column.

$$\therefore 15 = a_{32}$$

2. (a) $k = 3, p = n$

Explanation:

Order of matrix $P = p \times k$

Order of matrix $Y = 3 \times k$

Order of matrix $W = n \times 3$

$\therefore PY$ is defined if $k = 3$ and of order $p \times k$ and WY is defined and of order $n \times k$

$\therefore PY + WY$ is defined if $p = n$ and $k = 3$

3.

(b) $(AB)^t = B^t A^t$

Explanation:

By using the property of transpose, therefore $(AB)^t = B^t A^t$

4.

(b) $AB \neq BA$

Explanation:

Here, order of matrix A is 2×3 and order of matrix B is 3×2 So, AB will be of order 2×2 and BA will be of order 3×3 .

$$\text{Now, } AB = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 - 8 + 6 & 3 - 10 + 3 \\ -8 + 8 + 10 & -12 + 10 + 5 \end{bmatrix} \text{ [multiplying rows by columns]}$$

$$AB = \begin{bmatrix} 0 & -4 \\ 10 & 3 \end{bmatrix} \text{ and } BA = \begin{bmatrix} 2 & 3 \\ 4 & 5 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 & 3 \\ -4 & 2 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 2 - 12 & -4 + 6 & 6 + 15 \\ 4 - 20 & -8 + 10 & 12 + 25 \\ 2 - 4 & -4 + 2 & 6 + 5 \end{bmatrix} \text{ [multiplying rows by columns]}$$

$$= \begin{bmatrix} -10 & 2 & 21 \\ -16 & 2 & 37 \\ -2 & -2 & 11 \end{bmatrix}$$

$AB \neq BA$

$$5. \text{ (a) } \begin{vmatrix} p^5 & q\left(\frac{p^5-1}{p-1}\right) \\ 0 & 1 \end{vmatrix}$$

Explanation:

$$\begin{aligned}
A^2 &= \begin{vmatrix} p & q \\ 0 & 1 \end{vmatrix} \begin{vmatrix} p & q \\ 0 & 1 \end{vmatrix} \\
&= \begin{vmatrix} p^2 & pq+q \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} p^2 & q(p+1) \\ 0 & 1 \end{vmatrix} \\
A^3 &= \begin{vmatrix} p^3 & q(p^2+p+1) \\ 0 & 1 \end{vmatrix} \\
\text{Similarly } A^4 &= \begin{vmatrix} p^4 & q(p^3+p+1) \\ 0 & 1 \end{vmatrix} \\
\text{And } A^5 &= \begin{vmatrix} p^5 & q(p^4+p^2+p+1) \\ 0 & 1 \end{vmatrix} \\
\Rightarrow A^5 &= \begin{vmatrix} p^5 & q\left(\frac{p^5-1}{p-1}\right) \\ 0 & 1 \end{vmatrix} \quad [\because p^4+p^3+p^2+p+1 \text{ is in G.P. with } a=1 \text{ and } r=p]
\end{aligned}$$

6. (a) - (iii), (b) - (i), (c) - (iv), (d) - (ii)
7. (a) - (ii), (b) - (iii), (c) - (iv), (d) - (i)
8. (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)
9. (a) - (iii), (b) - (iv), (c) - (i), (d) - (ii)
10. (a) - (ii), (b) - (iv), (c) - (i), (d) - (iii)

11. $A^2 = A.A$

$$\begin{aligned}
&= \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} \\
&= \begin{bmatrix} 9-8 & -6+4 \\ 12-8 & -8+4 \end{bmatrix} \\
&= \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix}
\end{aligned}$$

$$A^2 = kA - 2I.$$

$$\begin{aligned}
\Rightarrow \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix} &= k \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\
\Rightarrow \begin{bmatrix} 1 & -2 \\ 4 & -4 \end{bmatrix} &= \begin{bmatrix} 3k & -2k \\ 4k & -2k \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\
\Rightarrow \begin{bmatrix} 3 & -2 \\ 4 & -2 \end{bmatrix} &= \begin{bmatrix} 3k & -2k \\ 4k & -2k \end{bmatrix} \\
\Rightarrow 3k &= 3 \\
\Rightarrow k &= 1
\end{aligned}$$

12. We have, $f(x) = x^2 - 4x + 7$

$$\therefore f(A) = A^2 - 4A + 7I_2$$

$$\begin{aligned}
\text{Now, } A^2 &= \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 4-3 & 6+6 \\ -2-2 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} \\
-4A &= 4 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} -8 & -12 \\ 4 & -8 \end{bmatrix} \text{ and, } 7I_2 = 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}
\end{aligned}$$

$$\therefore f(A) = A^2 - 4A + 7I_2$$

$$\begin{aligned}
\Rightarrow f(A) &= \begin{bmatrix} 1 & 12 \\ -4 & 1 \end{bmatrix} + \begin{bmatrix} -8 & -12 \\ 4 & -8 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} \\
\Rightarrow f(A) &= \begin{bmatrix} 1-8+7 & 12-12+0 \\ -4+4+0 & 1-8+7 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0_2
\end{aligned}$$

$$\text{Now, } f(A) = O_2$$

$$\Rightarrow A^2 - 4A + 7I_2 = O_2$$

$$\Rightarrow A^2 = 4A - 7I_2$$

By replacing the above equation in A^3 , we have

$$\Rightarrow A^3 = A^2 A = (4A - 7I_2) A = 4A^2 - 7I_2 A$$

$$\Rightarrow A^3 = 4(4A - 7I_2) - 7A \quad [\text{Using: } A^2 = 4A - 7I_2]$$

$$\Rightarrow A^3 = 9A - 28I_2$$

By successive replacements of A^3 we get,

$$\Rightarrow A^4 = A^3 A = (9A - 28I_2)A$$

$$\Rightarrow A^4 = 9A^2 - 28A = 9(4A - 7I_2) - 28A \text{ [Using : } A^2 = 4A - 7I_2]$$

$$\Rightarrow A^4 = 36A - 63I_2 - 28A = 8A - 63I_2$$

$$\Rightarrow A^5 = A^4 A = (8A - 63I_2)A = 8A^2 - 63I_2 A$$

$$\Rightarrow A^5 = 8(4A - 7I_2) - 63A = -31A - 56I_2 \text{ [Using : } A^2 = 4A - 7I_2]$$

$$\begin{aligned} \Rightarrow A^5 &= -31 \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} - 56 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} -62 & -93 \\ 31 & -62 \end{bmatrix} + \begin{bmatrix} -56 & 0 \\ 0 & -56 \end{bmatrix} = \begin{bmatrix} -118 & -93 \\ 31 & -118 \end{bmatrix} \end{aligned}$$

13. Let $A = [a_{ij}]_{4 \times 3}$

So, the elements in a 4×3 matrix are

$a_{11}, a_{12}, a_{13}, a_{21}, a_{22}, a_{23}, a_{31}, a_{32}, a_{33}, a_{41}, a_{42}, a_{43}$

These elements are written in the 4 rows and 3 columns matrix as given below.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{bmatrix} \dots\dots(1)$$

By successive replacements of the values of i and j we get the following elements of the given 4×3 matrix.

$$a_{11} = 2 \times 1 + \frac{1}{1} = 2 + 1 = 3$$

$$a_{12} = 2 \times 1 + \frac{1}{2} = 2 + \frac{1}{2} = \frac{5}{2}$$

$$a_{13} = 2 \times 1 + \frac{1}{3} = 2 + \frac{1}{3} = \frac{7}{3}$$

$$a_{21} = 2 \times 2 + \frac{2}{1} = 4 + 2 = 6$$

$$a_{22} = 2 \times 2 + \frac{2}{2} = 4 + 2 = 5$$

$$a_{23} = 2 \times 2 + \frac{2}{3} = 4 + \frac{2}{3} = \frac{14}{3}$$

$$a_{31} = 2 \times 3 + \frac{3}{1} = 6 + 3 = 9$$

$$a_{32} = 2 \times 3 + \frac{3}{2} = 6 + \frac{3}{2} = \frac{15}{2}$$

$$a_{33} = 2 \times 3 + \frac{3}{3} = 6 + 1 = 7$$

$$a_{41} = 2 \times 4 + \frac{4}{1} = 8 + 4 = 12$$

$$a_{42} = 2 \times 4 + \frac{4}{2} = 8 + 2 = 10$$

$$a_{43} = 2 \times 4 + \frac{4}{3} = 8 + \frac{4}{3} = \frac{28}{3}$$

So, from (1)

$$A = \begin{bmatrix} 3 & \frac{5}{2} & \frac{7}{3} \\ 6 & 5 & \frac{14}{3} \\ 9 & \frac{15}{2} & 7 \\ 12 & 10 & \frac{28}{3} \end{bmatrix}$$

$$14. 2X + 3Y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} \dots\dots(1)$$

$$3X + 2Y = \begin{bmatrix} 2 & -2 \\ -1 & 5 \end{bmatrix} \dots\dots(2)$$

Now, multiply equation (1) by 2 and equation (2) by 3, we get,

$$4X + 6Y = \begin{bmatrix} 4 & 6 \\ 8 & 0 \end{bmatrix} \dots\dots(3)$$

$$9X + 6Y = \begin{bmatrix} 6 & -6 \\ -3 & 15 \end{bmatrix} \dots\dots(4)$$

Subtracting equation (4) from (3), we get,

$$(4X + 6Y) - (9X + 6Y) = \begin{bmatrix} 4 & 6 \\ 8 & 0 \end{bmatrix} - \begin{bmatrix} 6 & -6 \\ -3 & 15 \end{bmatrix}$$

$$\Rightarrow -5X = \begin{bmatrix} 4-6 & 6-(-6) \\ 8-(-3) & 0-15 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 12 \\ 11 & -15 \end{bmatrix}$$

$$\Rightarrow X = -\frac{1}{5} \begin{bmatrix} -2 & 12 \\ 11 & -15 \end{bmatrix} = \begin{bmatrix} \frac{2}{5} & \frac{-12}{5} \\ \frac{-11}{5} & 3 \end{bmatrix}$$

$$\text{Now, } 2X + 3Y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix}$$

$$\Rightarrow 2 \begin{bmatrix} \frac{2}{5} & \frac{-12}{5} \\ \frac{-11}{5} & 3 \end{bmatrix} + 3Y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} \frac{4}{5} & \frac{-24}{5} \\ \frac{-22}{5} & 6 \end{bmatrix} + 3Y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix}$$

$$\Rightarrow 3Y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix} - \begin{bmatrix} \frac{4}{5} & \frac{-24}{5} \\ \frac{-22}{5} & 6 \end{bmatrix}$$

$$\Rightarrow 3Y = \begin{bmatrix} 2 - \frac{4}{5} & 3 + \frac{24}{5} \\ 4 + \frac{22}{5} & 0 - 6 \end{bmatrix} = \begin{bmatrix} \frac{6}{5} & \frac{39}{5} \\ \frac{42}{5} & -6 \end{bmatrix}$$

$$\Rightarrow Y = \frac{1}{3} \begin{bmatrix} \frac{6}{5} & \frac{39}{5} \\ \frac{42}{5} & -6 \end{bmatrix}$$

$$\Rightarrow Y = \begin{bmatrix} \frac{2}{5} & \frac{13}{5} \\ \frac{14}{5} & -2 \end{bmatrix}$$

15. We are given that,

$$\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$$

We need to find the value of b.

We know by the property of matrices, for the equality of matrices,

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

This implies,

$$a_{11} = b_{11}, a_{12} = b_{12}, a_{21} = b_{21} \text{ and } a_{22} = b_{22}$$

Similarly, by equating the corresponding entries of the two matrices on the both sides of the matrix equation, we get the following equations.

$$a - b = -1 \dots (i)$$

$$2a + c = 5 \dots (ii)$$

$$2a - b = 0 \dots (iii)$$

$$3c + d = 13 \dots (iv)$$

We have the equations (i), (ii), (iii) and (iv).

We need not solve equations (ii) and (iv). We will be able to solve for b from equations (i) and (iii).

Multiply equation (i) by 2.

$$a - b = -1 \quad [\times 2]$$

$$\Rightarrow 2a - 2b = -2 \dots (v)$$

Subtracting equation (iii) from equation (v),

$$2a - 2b = -2$$

$$2a - b = 0$$

$$\begin{array}{r} (-)(+)(-) \\ 0 - b = -2 \end{array}$$

$$\Rightarrow -b = -2$$

$$\Rightarrow b = 2$$

Thus, the value of b = 2.

16. We are Given a matrix , $A = \begin{bmatrix} 3 & 2 & 7 \\ 1 & 4 & 3 \\ -2 & 5 & 8 \end{bmatrix}$ Then $A^T = \begin{bmatrix} 3 & 1 & -2 \\ 2 & 4 & 5 \\ 7 & 3 & 8 \end{bmatrix}$

$$\begin{aligned} X &= \frac{1}{2} (A + A^T) \\ &= \frac{1}{2} \left(\begin{bmatrix} 3 & 2 & 7 \\ 1 & 4 & 3 \\ -2 & 5 & 8 \end{bmatrix} + \begin{bmatrix} 3 & 1 & -2 \\ 2 & 4 & 5 \\ 7 & 3 & 8 \end{bmatrix} \right) \\ &= \frac{1}{2} \begin{bmatrix} 3+3 & 2+1 & 7-2 \\ 1+2 & 4+4 & 3+5 \\ -2+7 & 5+3 & 8+8 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 6 & 3 & 5 \\ 3 & 8 & 8 \\ 5 & 8 & 16 \end{bmatrix} \\ X &= \begin{bmatrix} 3 & \frac{3}{2} & \frac{5}{2} \\ \frac{3}{2} & 4 & 4 \\ \frac{5}{2} & 4 & 8 \end{bmatrix} \end{aligned}$$

Now for matrix Y, we have

$$\begin{aligned} Y &= \frac{1}{2} (A - A^T) \\ &= \frac{1}{2} \left(\begin{bmatrix} 3 & 2 & 7 \\ 1 & 4 & 3 \\ -2 & 5 & 8 \end{bmatrix} - \begin{bmatrix} 3 & 1 & -2 \\ 2 & 4 & 5 \\ 7 & 3 & 8 \end{bmatrix} \right) \\ &= \frac{1}{2} \begin{bmatrix} 3-3 & 2-1 & 7+2 \\ 1-2 & 4-4 & 3-5 \\ -2-7 & 5-3 & 8-8 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 0 & 1 & 9 \\ -1 & 0 & -2 \\ -9 & 2 & 0 \end{bmatrix} \\ Y &= \begin{bmatrix} 0 & \frac{1}{2} & \frac{9}{2} \\ -\frac{1}{2} & 0 & -1 \\ -\frac{9}{2} & 1 & 0 \end{bmatrix} \end{aligned}$$

Now,

$$X^T = \begin{bmatrix} 3 & \frac{3}{2} & \frac{5}{2} \\ \frac{3}{2} & 4 & 4 \\ \frac{5}{2} & 4 & 8 \end{bmatrix}^T = \begin{bmatrix} 3 & \frac{3}{2} & \frac{5}{2} \\ \frac{3}{2} & 4 & 4 \\ \frac{5}{2} & 4 & 8 \end{bmatrix} = X$$

$\Rightarrow X$ is a symmetric matrix.

Now,

$$\begin{aligned} -Y^T &= - \begin{bmatrix} 0 & \frac{1}{2} & \frac{9}{2} \\ -\frac{1}{2} & 0 & -1 \\ -\frac{9}{2} & 1 & 0 \end{bmatrix}^T = - \begin{bmatrix} 0 & -\frac{1}{2} & -\frac{9}{2} \\ \frac{1}{2} & 0 & 1 \\ \frac{9}{2} & -1 & 0 \end{bmatrix} \\ -Y^T &= \begin{bmatrix} 0 & \frac{1}{2} & \frac{9}{2} \\ -\frac{1}{2} & 0 & -1 \\ -\frac{9}{2} & 1 & 0 \end{bmatrix} \end{aligned}$$

$$-Y^T = Y$$

Therefore Y is a skew symmetric matrix.

And,

$$X + Y = \begin{bmatrix} 3 & \frac{3}{2} & \frac{5}{2} \\ \frac{3}{2} & 4 & 4 \\ \frac{5}{2} & 4 & 8 \end{bmatrix} + \begin{bmatrix} 0 & \frac{1}{2} & \frac{9}{2} \\ -\frac{1}{2} & 0 & -1 \\ -\frac{9}{2} & 1 & 0 \end{bmatrix}$$

$$\begin{aligned}
&= \begin{bmatrix} 3+0 & \frac{3}{2} + \frac{1}{2} & \frac{5}{2} + \frac{9}{2} \\ \frac{3}{2} - \frac{1}{2} & 4+0 & 4-1 \\ \frac{5}{2} - \frac{9}{2} & 4+1 & 8+0 \end{bmatrix} \\
&= \begin{bmatrix} 3 & 2 & 7 \\ 1 & 4 & 3 \\ -2 & 5 & 8 \end{bmatrix} = A
\end{aligned}$$

Hence, $X + Y = A$

17. $|A| = 51$

Cofactors:

$$A_{11} = 28 \quad A_{12} = 13 \quad A_{13} = -19$$

$$A_{21} = -2 \quad A_{22} = 10 \quad A_{23} = 5$$

$$A_{31} = -17 \quad A_{32} = -17 \quad A_{33} = 17$$

$$A^{-1} = \frac{1}{51} \begin{bmatrix} 28 & -2 & -17 \\ 13 & 10 & -17 \\ -19 & 5 & 17 \end{bmatrix}$$

Given system is $AX = B$ where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ $B = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$

$$\Rightarrow X = A^{-1} B = \frac{1}{51} \begin{bmatrix} 28 & -2 & -17 \\ 13 & 10 & -17 \\ -19 & 5 & 17 \end{bmatrix} \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -2 \end{bmatrix}$$

$$x = 3, y = 2, z = -2$$

ATP